AUFIT
User's manual

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Presentation

This document describes the usage of the **AUFIT** program. AUFIT is the acronym for ***Atari Universal Floppy-Disk Image Tool***. The program can read many Floppy Disk image's formats used with the Atari platform including images generated at the flux level by devices like SuperCard Pro, KryoFlux, and Discovery Cartridge. Once read the FD image can be analyzed and the content can be displayed using Graphical and Text outputs. It is also possible to output the information read to several image formats. It is therefore possible to use the program to convert from one format to another. Usage has been designed to be as simple as possible for casual users. However the program also provides powerful features for advanced users.

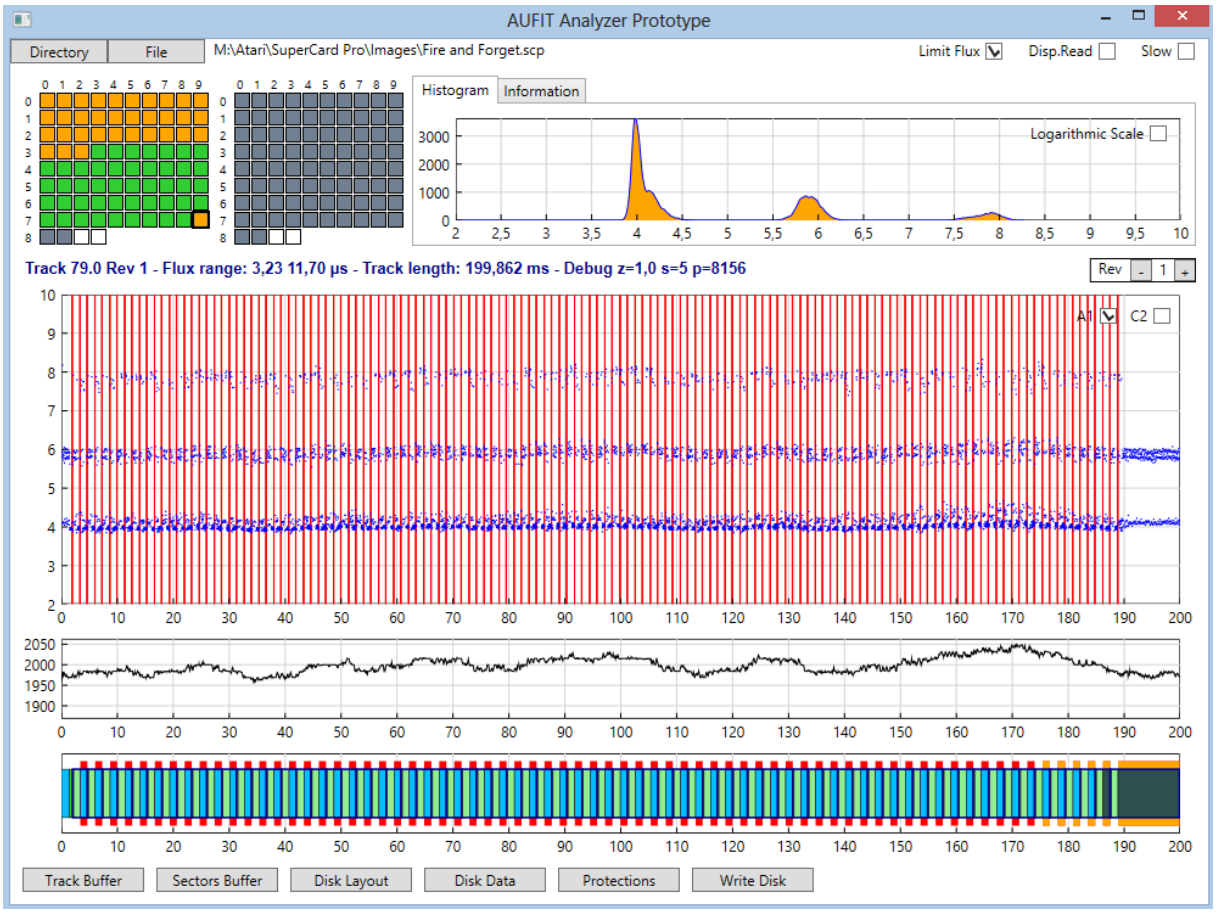
Release Note

- The program is still in alpha test. Several basic features are not yet implemented, the performance may be slow, and the program may terminate unexpectedly.
- Note: Some features of the program are documented feel free to experiment but at your own risk (do not worry the only danger is that the program crash).
- Sorry but currently most of the developments are focused on functionalities and therefore I know that the GUI needs a lot of rework ☺
- Currently the program uses a HUGE amount of memory. If you do not have a system with a lot of memory the program may run out of memory. Sometime restarting the program for looking at a new image may help.

Running the AUFIT Program

Currently the program does not requires to be installed and is provided directly as an executable: **Aufit.exe**. To run the program just double click the executable.

The program run under Windows Vista SP2, Windows 7 SP1, and Widows 8 or above. The program requires that the Microsoft.NET Framework 4.5 or above is installed. It also requires the Ookii Dialogs class library. The Ookii DLL is provided along with the executable and documentation.



Reading Image Files

Currently the program can read a complete Floppy disk image composed of a set of KryoFlux stream files, or a SuperCopy Pro file as well as a single track stream file.

To fully benefit from the ***Aufit*** analysis tools you need to sample at least **three** revolutions for each track and four is the recommended value. Please refer to the tools documentation to create images with 3 or more revolutions.

Reading Kryoflux FD set of stream files

The content of a complete floppy disk has to be saved as a set of raw stream files under a unique directory (the default using DTC). To read all the stream files in ***Aufit*** you must click the **Directory** Button. A form open and allow you to specify the location of the directory where all the files are located. The stream files inside the directory can have any name but of course they must have the same "root" name (e.g. awesome) followed by .Track_Num followed by .Side followed .raw. For

example awesome.00.0.raw, awesome.00.1.raw, awesome.01.0 etc. Reading all the files (usually 170 files) is IO intensive and may take some time.

✎ Two techniques are experimented: all files read simultaneously using multitasking (the default that seems to be the fastest) or files are read asynchronously displaying the progression (check the **Disp.Read** check box before reading).

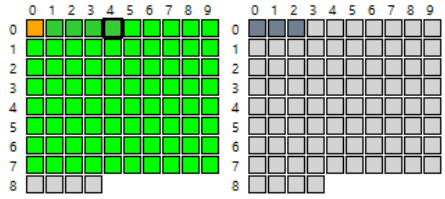
SuperCard Pro FD image

The image content of a complete floppy disk is saved by SuperCard Pro as one big file. For example Dungeon-Master.scp. To read the content of the Floppy disk image you must click the **File** button and select the file to open.

Kryoflux Single file

It is also possible to read a unique stream file using the **File** button. However beware that that several commands in Aufit only make sense on a complete disk image. For example displaying the Disk Layout.

Selecting the track to display



Once the FD image is loaded the information for the first track of the disk is displayed in different graphs. The program also displays two grids of selection buttons: one for each side of the floppy. At this stage only a quick analysis of the track content has been done and is used to determine the background color of the buttons: light green for normal tracks, light grey for unformatted tracks, and yellow for partially unformatted or "noisy" tracks.

As soon as a track is fully analyzed the background color will turn to: dark green for normal tracks, dark grey for unformatted/empty tracks, and red for protected tracks.

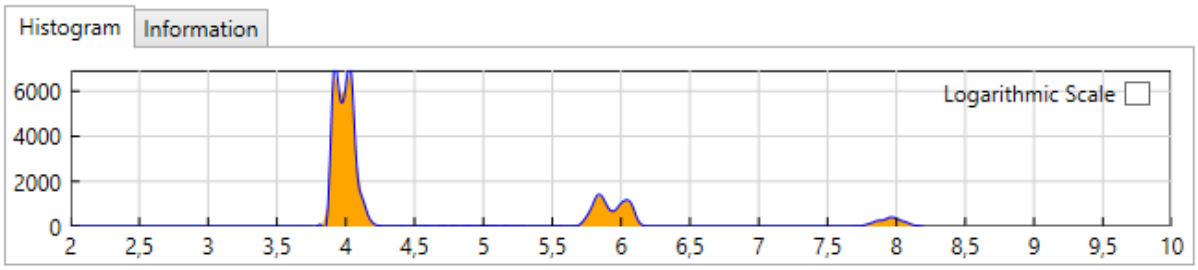
The currently selected track button can be identified by larger black border around the button. Leaving the mouse over a button display the track number.

If you click on a button of the grid this will cause the track to be fully analyzed and all the graph to be updated for the selected track. The background color is therefore updated as explained above. Once the selection grid is "active" (for example by clicking one button) you can move from one button to its neighbor using the direction keys on your keyboard.

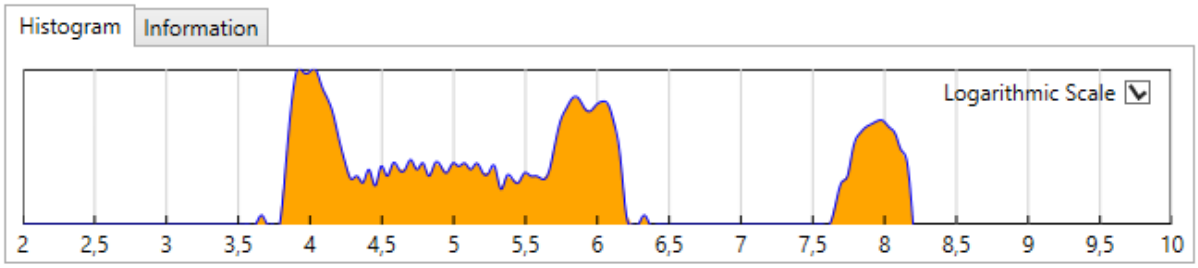
Note that full analysis of tracks is done whenever information about a track needs to be displayed. Therefore the **Disk layout**, **Disk Data**, **Protections**, and **Write Disk** buttons will force the full analysis of all tracks. Once analyzed a track does not need to be reanalyzed (analysis information is saved) and therefore many operations execute faster once all the tracks have been analyzed (for example **Disk Layout**).

The Histogram Graph

This histogram representing the flux transition repartition and is displayed at the right of the button grid. Normally you should see three pics at 4, 6, and 8 microseconds.

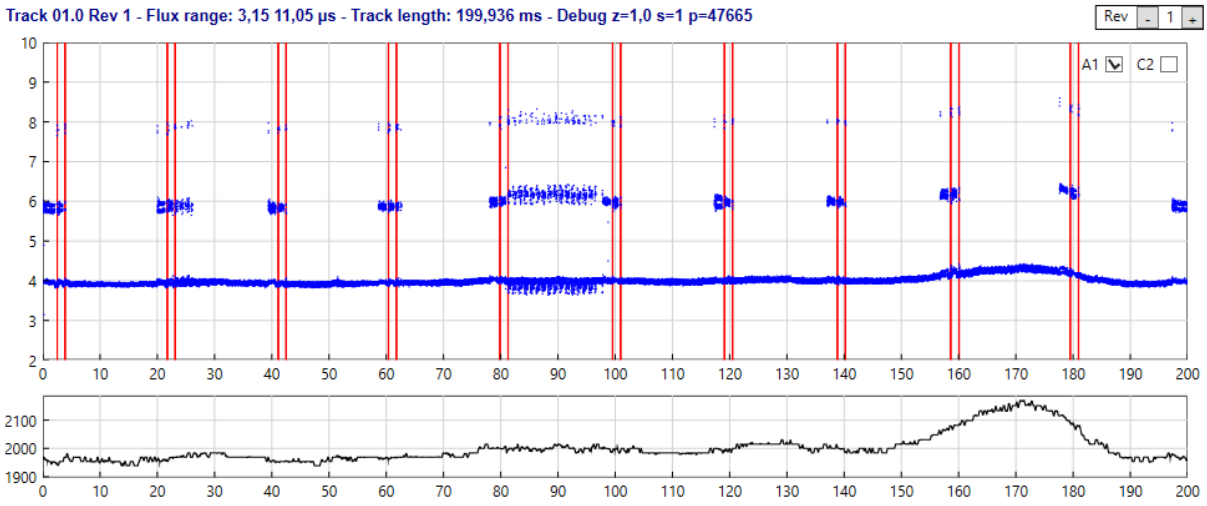


It is sometimes interesting to click the "**Logarithmic Scale**" check box to switch to a logarithmic scale on the Y axis. This allow to see information not visible on a linear scale. For example with DM you will see that the histogram has a lot of entries between 4 and 6 microsecond due to the "sliding fluxes" pattern used.



Flux Transitions Chart

The unprocessed flux transitions (i.e. flux directly read from the file) are displayed in a scattered chart. A standard track is about 40000 flux. To speed up the display time only about 10000 flux transitions are displayed by default. If you want to display all the flux transitions you can click the **slow** check box in upper right corner of the window, this will force all the transitions to be displayed but the display will be slow. In the upper right of this chart you have two check box called **A1** & **C2**. These check boxes are used to enable/disable the display of the 0xA1 and 0xC2 Sync Mark respectively as they would be detected by the WD1772 using a ***read track*** command (i.e. with the sync detector activated at all time). You have to refresh the display to see/hide the sync marks.



There is a check box button in the upper right corner of the program window called **Limit Flux**. By default this button is checked for unformatted track and is unchecked for formatted track but you can change its state at all time. Both the Histogram chart and the Flux Chart will be updated. This is very useful when the track contains long flux transitions (like with NFA) to keep the graph readable.

Information about the currently selected track are displayed in a text box with the following information: track number, side, revolution, min and max value of the flux transition range, the track length, and some debug info.

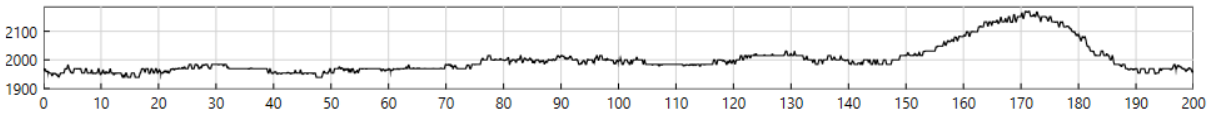
The box on the right indicates the currently used revolution. You can use the **+** and **-** button to select among the different revolutions sampled.

One very nice feature is that if you set the revolution number to 0 then the samples for **all** revolutions sampled are displayed! The transitions from different revolutions are separated by black lines.

This feature is nice to see protections like data over index, data beyond index, ID over index (and even Sync over index!). Be warned that if you check the **slow** button while displaying the data for all revolution the display will be very very slow as about 200000 points needs to be displayed.

Clock Chart

The clock chart displays the period of the internal DPLL clock. The nominal period is 2000 ns but of course can vary a lot for over/under clocked sectors tracks.



This clock is computed by the DPLL and represent a running average frequency of the flux transitions fed to the DPLL. The Y scale is automatically adjusted.

The clock should normally stay stable around the 2000ns period but for protection like Copylock the period usually goes from 2000 to 2100 (5% increase) inside a sector. Some protections over clock or under clock the complete track.

In some specific cases like unformatted track/sector the period can go up to 2200ns or down to 1800ns (10% limits preset in the DPLL).

Note that this graph uses processed (by the DPLL code) flux transitions.

Data Graph

This graph display the data as they would be read by the read track command of the WD1772. The following color are used:

- The ID field is displayed in yellow (you probably need to zoom to see it),
- the intra-gap is displayed in light green,
- the Data field is displayed in sky blue, and
- The inter-GAP is displayed in lime green.

The data field color can change to aqua (lighter blue) for under clocked sector and dark blue for overclocked sectors and blue violet for sector with intra byte bit width variation (e.g. speedlock).

Data sectors with CRC error have a red border, and data sectors with fuzzy bits have an orange border (this also imply CRC error).

If you place the cursor over a sector detailed information about the sector is displayed. If you double click a sector the content of this sector (as read by the FDC with a ***read sector*** command) is displayed in a window.

The display of the track data is processed by a complex algorithm (i.e. not a simple representation of a ***read track*** command). This permits to display correctly "strange" tracks containing protection like: Sector within sector, tracks with 70 sectors, tracks with one sector.

Note that No flux Area should be displayed correctly even if located over the index.

Zooming and Panning in the Charts

✎ This functionality has been implemented to provide Pan and Zoom quickly but the use model needs to be reworked.

When the flux chart is not zoomed you can use the left button to define a zoom area (both in x and y directions). Click on the first point to start the definition of the area, keep the mouse button down and move to the second point to define the zoom area. A rubber band box is drawn and when the left button is released the graph is zoomed in the defined area. The scale used on both axis is automatically adjusted by the zoom factor ☺

Once zoomed a click with the left button allow to pan the graph. Click any point in the flux graph with the left button and while keeping the button down move the cursor. The image of the graph dynamically moves with the cursor. When you release the mouse button, the graph and the axis position are updated.

You can further zoom-in or zoom-out by using the mouse wheel.

If you want to reset the graphs to un-zoomed state click anywhere in the Flux chart with the right mouse button (this is also the way to force a redisplay).

Note that the same zoom factor is used to update the flux chart, the clock chart, and the data chart. Therefore these three graphs always stay in synchronized.

✎ Note that zooming in the Y direction only happen in the flux chart.

✎ Note that you must place the mouse in the flux chart to pan and zoom

Track Buffer

If you click the **track buffer** button a new window will open that display several information about the currently selected track.

Track info

Contains the track/side number, the number of bytes, the track length as sampled by the imaging tool, and the number of sectors.

Track layout info

An array containing one line for each sector of a track, as well as a pre first sector that contains the GAP and eventually DATA segment information.

For each sector the following information is available:

- ID field: The sector ID, the position of the ID in microseconds from the index, the length of the ID field in microseconds and if the CRC of the ID is correct.
- Inter-Gap field: The number of bytes and the length of the gap in microseconds.
- Data Field: The number of bytes as decoded from the ID segment, the position in microseconds for the index, the length in microseconds, two indicators to display the status of the CRC and if the sector contains fuzzy bytes.
- Intra-Gap field: the number of bytes to reach the next sector and the length in microseconds.

Track content

A dump of all the data bytes exactly as they would have been read by the ***read track command*** of the WD1772. The data bytes are displayed by group of 16 with the following information:

- The first number is the position of the byte in the buffer (decimal value)
- The second number is the period of the first byte of the line.
- The third number is the position of the first byte of the line in microsecond relative to the index of the track.
- 16 bytes are displayed in hexadecimal
- The corresponding 16 ASCII characters for the 16 bytes.

Colors and font style are used to provide additional information.

Background color:

- Red for invalid characters (F5-F7) in gap
- Coral for no flux bytes (MFM 0x0000)
- Yellow for Sync characters (MFM 0x4489 and 0x5224)
- Light green for bytes inside a ID field or a Data field.

Character color:

- Red for 0x4489 Sync characters
- Green for 0x5224Sync characters
- Blue for ID bytes
- Black for Data and intra-gap byte
- Green for inter-gap byte

Bold characters:

- For Sync characters (MFM 0x4489 and 0x5224)

Sector Buffer

If you click the **sector buffer** button a new window will open that display the bytes exactly as it would have been read by combining the ***read address*** and ***read sector*** commands of the WD1772.

Sector info

Contains the track/side number, and the number of sectors.

Sector layout info

An array containing one line for each sector of a track.

For each sector the following information is available:

- ID field: The sector ID, the position of the ID in microseconds from the index, the length of the ID field in microseconds and if the CRC of the ID is correct.
- Data Field: The number of bytes as decoded from the ID segment, the position in microseconds for the index, the length in microseconds, two indicators to display the status of the CRC and if the sector contains fuzzy bytes.

Sectors content

For each sector a line that indicates the sector number, the number of bytes for this sector, the position in microseconds from the index, the length in microseconds, indications to say if the CRC is good or bad, and if the sector has fuzzy bytes.

This is followed by a dump of all the data bytes exactly as they would have been read by the ***read sector*** command of the WD1772. The data bytes are displayed by group of 16 with the following information:

- The first number is the position of the byte in the buffer (decimal value)
- The second number is the period of the first byte of the line.
- The third number is the position of the first byte of the line in microsecond relative to the index of the track.
- 16 bytes are displayed in hexadecimal
- The corresponding 16 ASCII characters for the 16 bytes.

If fuzzy bytes have been found for one sector a "fuzzy mask" buffer is also displayed.

To find fuzzy bytes the Aufit program read the same sector multiple times from the different revolutions. If the content of a byte in the same position is different the mask value is the result of a **xor** operation.

```
Mask[offset] = (Sector_read_1[offset] ^ Sector_read_2[offset])
```

Thus same value (1&1 or 0&0) gives a result of zero and different value (0&1 or 1&0) gives a result of one. For example if the same byte is read one time as 0000 1111 and the next time as 0101 0101 the mask will be 0101 1010.

The mask can then be used to emulate fuzzy bits as follow:

```
FuzzyByte[offset] = (Sector[offset] & Mask[offset]) | (rand() & Mask[offset])
```

Where rand() is a function that randomly returns true or false.

The information for a sector presented above repeats for all sectors.

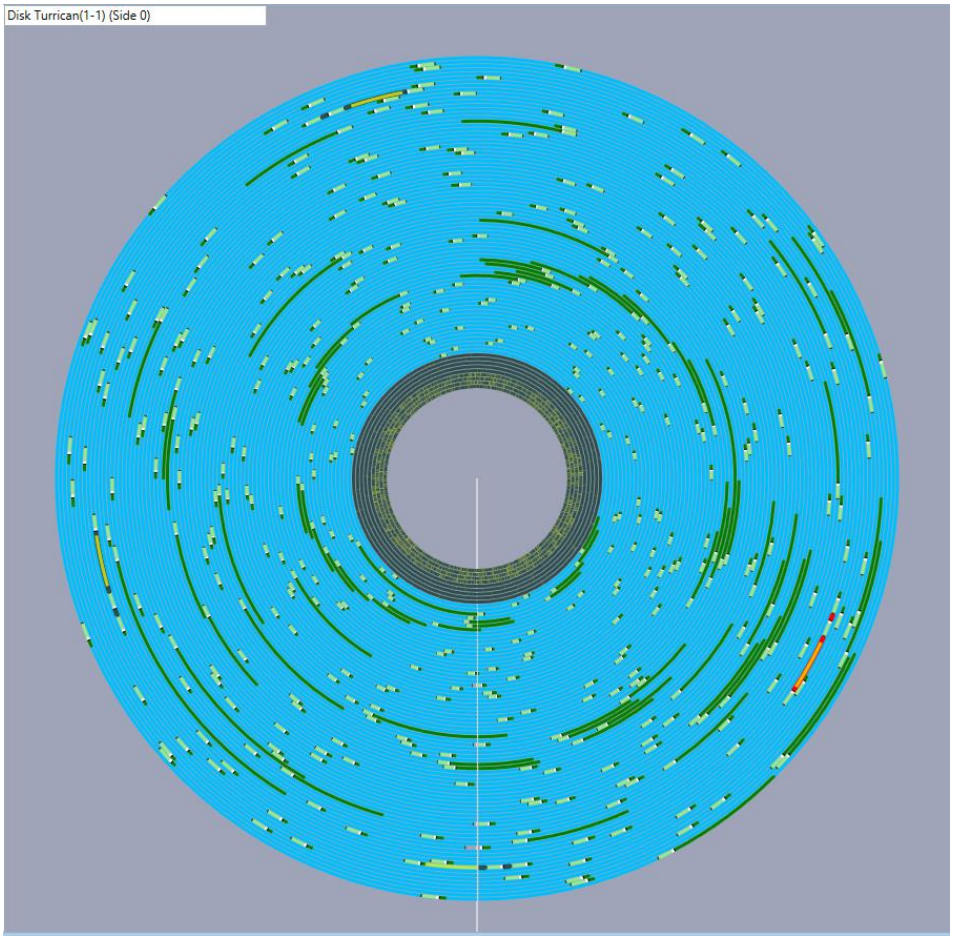
Disk Layout

If you click the **Disk Layout** button a new window will open to display the layout of one side of the floppy disk. The side is determined by the currently selected track. This display requires to analyze each and every tracks of the displayed side of the Floppy disk. Therefore, unless the track are already analyzed, this operation can take a long time to execute. During this operation the interface do not freeze and it is therefore possible to still select and display different tracks in the main window..

The following color information is used:

- The ID field of a sector is in white
- The Intra-gap of a sector is in light green
- The Data field of a sector is displayed:
  - In Deep sky blue by default
  - In blue for over clocked sector
  - In Aqua for under clocked segment
  - In Violet for sector with intra byte speed variation (e.g. speedlock)
- The Inter-gap are displayed in green
- Data sector with CRC error or Fuzzy bytes are displayed in red
- No Flux region are displayed in Yellow
- Sync Character are displayed in Black

For example the Turrican Game that has many protection and an irregular layout is displayed like this.



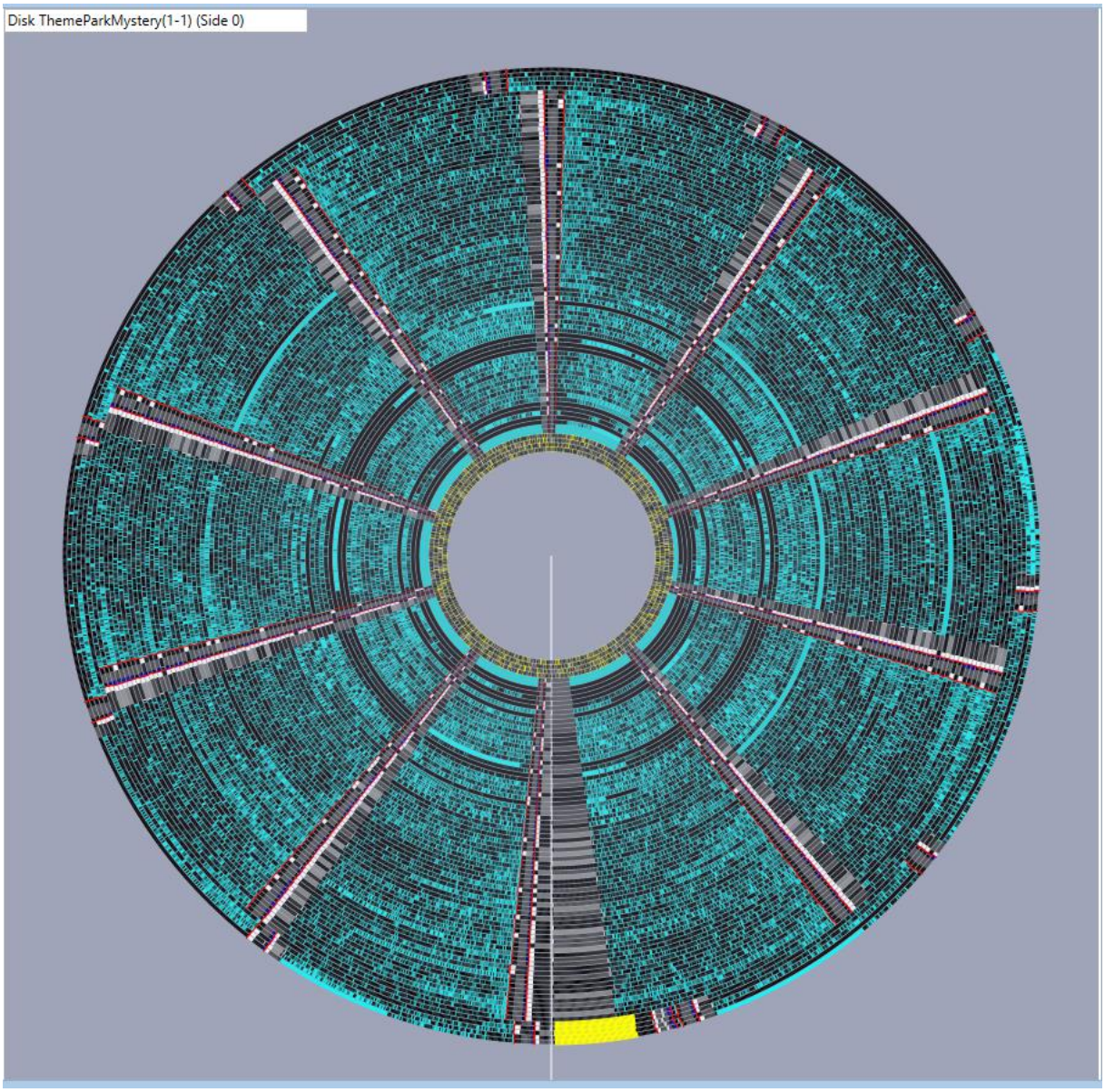
Disk Data

If you click the **Disk Data** button a new window will open and display the data on the floppy. This graph displays a lot of information so be patient.

The following color are used

- The Gap data are displayed in grey color proportional to the data value (so 0x00 is black and 0xFF is white and 0x55 is displayed as RGB={55,55,55}
- The Data uses the same principle but only using two color channels. For example 0x55 is displayed using RGB={00,55,55}
- The ID uses the blue channel for example RGB={00,00,55}
- NFA are displayed using yellow color
- Sync in Red

Here is an example of Disk Data Graph on the Theme Park Mystery Game



Protections

Protections are analyzed when a Track is displayed. You can use the **Protection** button to force the analysis of all the tracks. When a track is fully analyzed the background color in the track grid is updated as explained in Selecting the track to display. The **Disk Layout** and **Disk Data** button will also force the analysis of all the tracks.

You can select the **Information** tab close to the histogram to display protection information. If you use the protection button all the protections for all the tracks are displayed in the information window.

If you change the currently displayed track by clicking a track in the grid or by using the keyboard direction keys, all the protections for the selected track are displayed.

The following protections are currently analyzed.

- Track Protections
  - Short Track
  - Long Track
  - Track has No Flux Area
  - Track has data over index
  - Track has data beyond index
  - Track has id over index
  - Missing track
  - Extra track
- Sector Protections
  - Sector not in sequence 1..n (For Pasti)
  - Duplicated Sectors
  - Sector has invalid number
  - Sector has nonstandard size (for Pasti)
  - Sector has Invalid data into Gap
  - Sector has invalid sync sequence
  - Sector Data has CRC error
  - Sector ID has CRC error
  - Sector has fuzzy bytes
  - Long sector
  - Short sector
  - Intra sector bit width variation (e.g. speedlock)
  - Sector has no data
  - Sector with invalid track number
  - Sector contains sector

For more details about these protections please refer to my document **Atari Floppy Disk Copy Protection**.

Document History

- V0.1a - 9 January 2014 – Preliminary release for v0.1 Alpha program.
- V0.0 – 6 January preliminary first release.

Program History

- V0.1a January 9, 2014 – Compared to initial release Protection analysis has been added. So only missing initial features is Pasti write capability.
  Known problems: Run out of memory easily, GUI freeze with DISK DATA command.
- V0.0 January 6, 2014 – Initial alpha release on request.